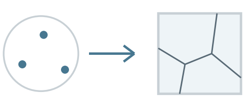
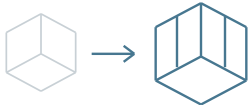
COMPLETED AND PRELIMINARY FOUNDATIONS



COMPLETED

Control simulation physics

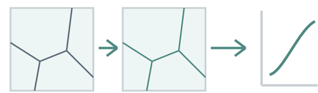
Neighborhood-defined labels



COMPLETED - UNDER REVIEW

Scale local learning

Scalable 3D-PRIMME rollouts



AVAILABLE EXPERIMENT

Measure 4D evolution

One five-state DCT trajectory

EXPERIMENT-CENTERED AIMS

Measured 4D grain evolution is the scientific target

Local learning is the tool; experimental behavior defines the claims

AIM 1

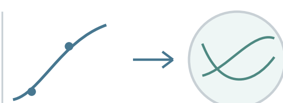
Select, curate, and train one 4D window

Selection provenance, label integrity, and drift control

AIM 2

Evaluate rollout on later measured states

Retrospective T2-T4 fidelity and simple baseline comparisons



AIM 3

Interpretability and physics learning

Supported responses advance toward bounded physical attribution